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WTU-EVAL: A Whether-or-Not Tool Usage Evaluation Benchmark for Large Language Models

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***Abstract*—Although Large Language Models (LLMs) excel in NLP tasks, they still need external tools to extend their ability. Current research on tool learning with LLMs often assumes mandatory tool use, which does not always align with real-world situations, where the necessity for tools is uncertain, and incorrect or unnecessary use of tools can damage the general abilities of LLMs. Therefore, we propose to explore whether LLMs can discern their ability boundaries and use tools flexibly. We then introduce the Whether-or-not tool usage Evaluation benchmark (WTU-Eval) to assess LLMs with eleven datasets, where six of them are tool-usage datasets, and five are general datasets. LLMs are prompted to use tools according to their needs. The results of eight LLMs on WTU-Eval reveal that LLMs frequently struggle to determine tool use in general datasets, and LLMs' performance in tool-usage datasets improves when their ability is similar to ChatGPT.**

***Index Terms*—Large Language Models, Tool Usage, Evaluation, Ability Boundary**

I. Introduction

LLMs like GPT4 [26] excel in general-purpose NLP tasks, which however struggle in tasks requiring calling external tools, such as fetching up-to-date stock prices and booking flights [1], [2], [19], [20]. How to improve LLMs' interaction with tools is a current hot topic. Recent studies have explored tool usage fine-tuning [2], [19] and reinforcement learning [27] techniques, showing promising results in areas like math reasoning and web

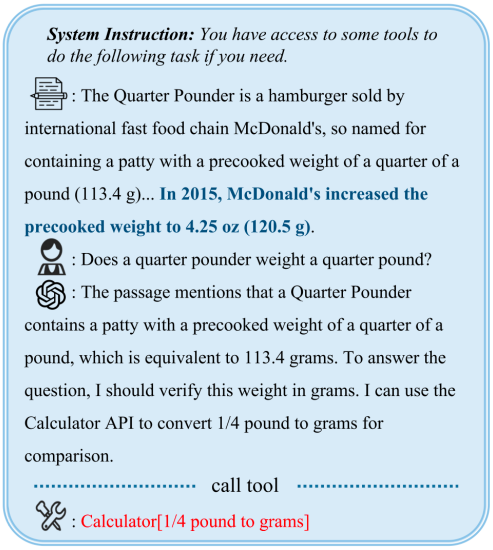

Fig. 1: An example showing the failure of calling tools inappropriately.

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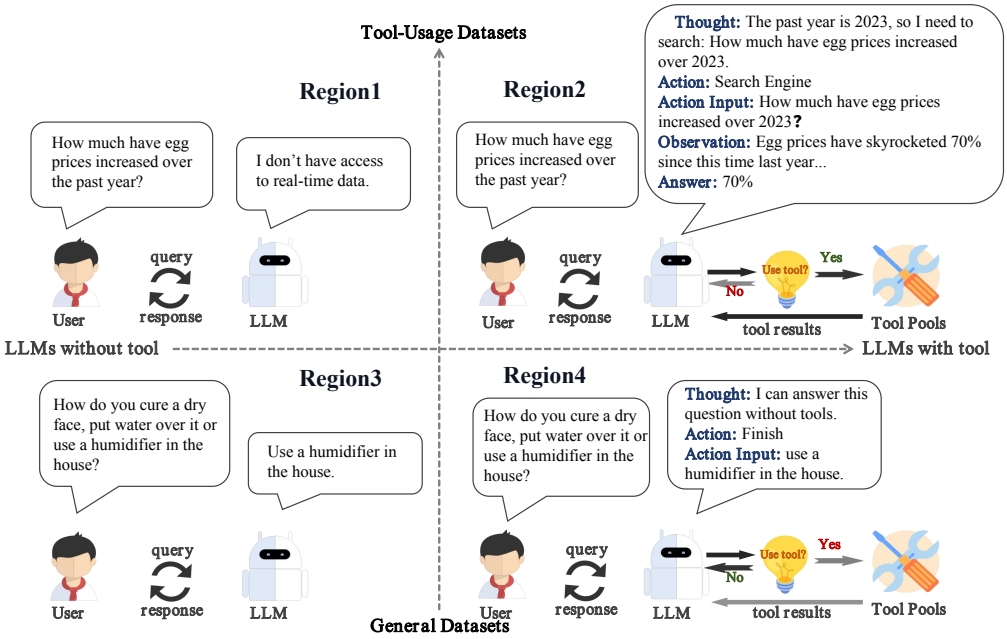

Fig. 2: Illustrative diagram depicting user interaction scenarios with and without access to tool pools. LLMs need to respond to the user's query in *Region1* (R1) and *Region3* (R3). In *Region2* (R2) and *Region4* (R4), LLMs must judge based on the nature of the task whether a tool is required. If so, the corresponding tool from the tool pool is invoked; if not, the answer is provided using its knowledge. If the judgment is correct, then the corresponding choice is highlighted in green; otherwise, it is in red.

search.

Despite notable progress, prior studies [3], [20] mainly focused on scenarios mandating tool use by LLMs. However, in a real-world application, the necessity for tool usage is uncertain. Moreover, we observe that inappropriate tool invocation can lead to errors, adversely affecting outcomes. For example, Figure 1 provides an example of using ChatGPT (0613) to answer a question. Despite the context hinting at the answer: a quarter pounder's weight has been increased to 120.5g, not a quarter pound (113.4g), ChatGPT still invokes an external tool, *Calculator*, and due to incorrect parameter settings, it produces an erroneous response and redundant response time.

With the above observations, we want to explore an intriguing question: whether LLMs can discern their ability boundaries, and if LLMs have the option to decide whether to use tools, would their performance improve in general and tool-usage datasets?

To this end, we propose a Whether-or-not tool usage Evaluation benchmark (WTU-Eval), which contains six tool-usage datasets that explicitly require tool usage and five general datasets that can be answered without tools.

II. The WTU-Eval Benchmark

The desired diagram of WTU-Eval is shown in Figure 2. In R1, the user asks a real-time question, but LLM cannot access this information without the search engine, so it fails to answer. In R2, when faced with the same question, LLM has access to tool pools and knows that the tool usage is necessary, so it decides to call *Search Engine* to find the real-time information and gives the correct answer. In R3, the user asks a general question, and LLM answers it with its knowledge. In R4, when presented with the same question, the LLM can access tool pools. Recognizing that tool usage is unnecessary, it decides to provide an answer directly.

By comparing the results between R1 and R2, we can determine whether LLMs recognize when a question exceeds their capabilities and thus requires the use of tools,

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TABLE I: The accuracy of experiments executed general datasets and tool datasets whether or not have access to tool pools, where "T003" means "Text-Davinci-003", and "" indicates few-shot results, while cells without background color indicate zero-shot results.

Test Set	Model w/o Tool						Model w/ Tool					
	T003	ChatGPT	Llama2-13B		Llama2-7B		T003	ChatGPT	Llama2-13B		Llama2-7B	
			Base	Chat	Base	Chat			Base	Chat	Base	Chat
Tool datasets	**R1**						**R2**					
MLQA	54.17	53.13	**52.08**	57.29	**55.21**	**62.50**	58.33	50.00	0.00	12.50	1.04	11.45
							70.83	**65.62**	26.04	**60.41**	50.00	48.95
ASDiv	48.67	79.33	**13.00**	**50.00**	23.00	45.67	**70.66**	**83.00**	9.00	23.66	**46.66**	38.66
							68.33	**83.00**	9.00	45.00	43.00	**47.66**
GSM8K	14.00	67.00	**2.00**	9.00	9.00	**12.00**	39.00	**58.00**	0.00	**20.00**	2.00	8.00
							52.00	53.00	0.00	15.00	**14.00**	5.00
MathQA	33.00	18.00	**12.00**	**17.00**	**19.00**	**26.00**	37.00	**39.00**	8.00	11.00	4.00	11.00
							39.00	36.00	6.00	12.00	10.00	5.00
RealtimeQA	36.66	**40.00**	**20.00**	30.00	23.34	40.00	**56.66**	36.66	0.00	23.33	3.30	40.00
							36.66	40.00	0.00	**33.33**	**26.66**	**26.66**
HotPotQA	33.50	34.50	**11.50**	**33.00**	**20.00**	**36.00**	28.50	39.00	0.00	18.00	0.00	20.50
							47.95	**41.50**	5.00	26.50	18.50	20.50
General datasets	**R3**						**R4**					
BoolQ	**79.00**	**89.00**	**56.00**	46.00	46.00	**57.00**	20.00	6.00	0.00	0.00	0.00	2.00
							58.50	76.25	54.00	**61.25**	**55.00**	32.50
RACE	68.96	**79.09**	14.80	22.93	32.00	33.87	6.00	30.00	0.00	0.00	14.00	0.00
							82.93	77.46	**62.05**	**52.53**	**58.40**	**50.40**
PIQA	**58.00**	**84.00**	16.00	32.00	**25.00**	**49.00**	3.00	39.00	0.00	0.00	0.00	0.00
							50.25	58.75	**27.25**	**43.50**	18.75	31.00
RTE	59.00	**78.00**	**68.00**	**63.00**	**54.00**	**58.00**	3.00	12.00	0.00	0.00	0.00	0.00
							66.25	50.00	13.50	34.50	45.50	23.25
HellaSwag	**54.00**	**75.00**	**21.00**	**49.00**	**54.00**	**44.00**	23.00	28.00	0.00	0.00	1.00	0.00
							50.75	50.00	4.25	23.25	20.00	15.75

and quantify the impact of using tools. By comparing the results of R3 and R4, we can determine whether the LLMs, when given the option to use tools, recognize that the current question can be answered without tools. Additionally, we can quantify the damage when they choose to use tools unnecessarily.

A. Evaluation Settings

a) Tool Pools: Following BMTools [1], we select the tools used in the evaluation, where machine translator and calculator are single-action tools, and search engine and Wikipedia search are multiple-action tools.

b) Datasets: We partition the datasets into the tool datasets (for tasks requiring specific tools), and the general datasets (for tasks solvable with LLMs' own ability). The tool datasets include MLQA [10], ASDiv [13], GSM8K [8], MathQA [9], HotpotQA [11], and RealtimeQA [12], focusing on machine translation, math reasoning, Wikipedia search, and search engine. The general datasets contain BoolQ [14], RACE [15], PIQA [16], RTE [21], and HellaSwag [17], focusing on reading comprehension, commonsense reasoning, and sentence completion. Test sizes are chosen based on reasoning complexity and time costs, leading to each random sampling of the size from 100 to 300 from original test sets.

c) LLMs: We test LLMs from both commercial and open-source sectors for a broad evaluation, including Text-Davinci-003, ChatGPT (0613), and Llama2 [22].

d) Evaluation Metrics: In WTU-Eval, we prioritize accuracy using advanced methods beyond exact matches, categorizing datasets into numerical and free-text responses. We check numerical answers with specific data and transform free-text responses into labels.

Additionally, tool usage is marked incorrect in the

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TABLE II: Accuracy in general datasets without tool access in COT and zero-shot settings.

Test set	ChatGPT		Llama2-7B		Llama2-7B-Chat	
	Zero-shot	COT	Zero-shot	COT	Zero-shot	COT
BoolQ	**89.00**	81.00	46.00	**74.00**	**57.00**	51.00
RACE	79.09	**83.47**	60.30	**65.07**	33.87	**67.73**
PIQA	84.00	**86.00**	25.00	**56.00**	49.00	**54.00**
RTE	78.00	78.00	54.00	**57.00**	**58.00**	47.00
HellaSwag	**75.00**	66.00	**54.00**	24.00	**44.00**	39.00

general dataset and a correct example is shown in Figure 2 R4.

B. Evaluation Prompt

In WTU-Eval, we utilize ReACT [18] for zero-shot and few-shot experiments in scenarios with access to tool pools (R2 and R4). The ReACT is structured into four stages: *Thought*, *Action*, *Observation*, and *Final Answer*, performed in a limited loop. Figure 1 also shows the system instruction we design in all experiments. To ensure fairness, all LLMs are evaluated under the same settings during the assessment.

a) Zero-shots: We introduce tool names, descriptions, and parameters to guide the LLMs to use tools.

b) Few-shots: We introduce tool names, descriptions, parameters, and examples of: a) one tool usage scenario, and b) one general scenario where tools are not used.

III. Experiments and Results

A. LLMs' performance in tool datasets

When LLMs are given the option not to use tools, both ChatGPT and Text-Davinci-003 demonstrate improved performance in tool-using scenarios, whereas Llama2 exhibits bad performance even with tool-using demonstration. In Table I R2, when LLMs have access to tools, Llama2-13B's performance on most tool tasks drops to 0, while ChatGPT and Text-Davinci-003 exhibit significant improvements (by up to 25% in GSM8K), exceeding their performance in R1. It is observed that the use of tools does not unconditionally enhance LLMs' performance and the enhancement depends on LLMs' ability. This primarily stems from the increased complexity of tool instruction, leading to models that use tools often performing worse compared to their zero-shot performance in R1.

Nevertheless, this trend alters with the adoption of the few-shot setting. In R2, ChatGPT and Text-Davinci-003's performance also improve (by up to 40% in GSM8K), exceeding their performance in R1. In contrast, Llama2 only shows improvement on a small portion of tool datasets, with performance declining on the rest compared to R1.

B. LLMs' performance in general datasets

LLMs do not know their ability limits clearly and call the tool to get answers in most general scenarios. Compared R4 to R3 in Table I, most LLMs' performance decreases in all general datasets. Notably, the most substantial decrease observed is 83% in BoolQ (Text-Davinci-003), and Llama2's performance nearly falls to 0. This is primarily due to LLMs' frequent misuse of tools in general queries, which results in an increased error rate. Error analysis also suggests that the complexity of following tool invocation instructions complicates adherence to the ReACT framework, impacting the *Thought* process.

C. Ablation Study

To mitigate the challenge of difficult tool instruction for LLMs without tuning with tool instruction, we conduct few-shot experiments in R4. These experiments largely echo the zero-shot findings, with slight improvements observed when tools are unavailable (R3). Importantly, this slight increase is mainly observed in Llama2, which shows a 10% to 30% improvement. This improvement is likely due to the integration of a Chain of Thought (COT) to solve the task and specific response formats.

To explore the impact of the COT and ReACT's format, we conduct few-shot trials without tool access (R3), focusing on the COT process. The results, as shown in Table II, reveal that COT significantly aids smaller-scale Llama models. However, for ChatGPT, COT does not lead to improvements and might even result in performance declines on BoolQ and HellaSwag.

IV. Conclusion

In this paper, we explore whether LLMs can discern their ability boundaries and use tools flexibly. We introduce the WTU-Eval to assess LLMs with eleven datasets and four tools, which is the first evaluation to evaluate whether to use tools accurately. The results reveal that LLMs frequently struggle to determine tool use in general datasets, and their performance in tool-usage datasets improves when their ability is similar to ChatGPT. In both datasets, incorrect tool usage significantly impairs LLMs' performance.

Our work points out the overlooked shortcomings in tool usage by LLMs, i.e., they struggle to recognize their capability boundaries. Future works include adding more datasets and tools and testing more types of LLMs.

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